

## Review

# Comparative Assessment of LEED, BREEAM, and WELL: Advancing Sustainable Built Environments

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## Abstract

This study compares the LEED, BREEAM, and WELL certification systems using the Triple Bottom Line (TBL) framework to assess their performance across environmental, social, and economic dimensions and their alignment with sustainable development goals. A structured secondary analysis was conducted on over 50 peer-reviewed articles, case studies, and official certification manuals. Inclusion criteria required documented design targets and post-occupancy outcomes for certified buildings (2014–2024). A two-phase analytical model was applied: first, evaluating each system's structure and priorities; then benchmarking them using the TBL framework to assess how holistically each addresses sustainability. Results show that LEED leads to energy optimization, BREEAM to lifecycle integration, and WELL to occupant health and indoor environmental quality. However, all systems exhibit post-occupancy performance gaps: LEED and BREEAM underperform by 15–30% in energy use, while WELL-certified projects may exceed 30% due to stringent indoor comfort demands. These findings highlight the need to integrate real-time post-occupancy evaluation into certification protocols. To improve overall effectiveness, the study proposes enhancements such as adaptive performance tracking, occupant feedback loops, and dynamic benchmarking aligned with actual building use. By identifying both the comparative strengths and systemic limitations of the three frameworks, this research contributes to the refinement of green building assessment tools. Practical implications include (1) integrating post-occupancy evaluation into certification renewal cycles, (2) adopting hybrid certification strategies to improve sustainability coverage, and (3) designing benchmarking tools that reflect real-world operational data.

**Keywords:** LEED; BREEAM; WELL Building Standard; sustainability certification; Triple Bottom Line (TBL); energy efficiency; carbon reduction; green building



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## 1. Introduction

The global building sector accounts for over 34% of global energy demand and 37% of energy and process-related CO<sub>2</sub> emissions [1]. Due to its environmental footprint, the sector plays a pivotal role in achieving climate targets and sustainable development. Green building certification systems (GBCSs) have become essential tools for driving more sustainable practices across environmental, economic, and social domains.

Among the most widely adopted GBCSs are LEED (Leadership in Energy and Environmental Design), BREEAM (Building Research Establishment Environmental Assessment

Method), and the WELL Building Standard. These systems guide sustainable design, construction, and operations through defined criteria and performance benchmarks.

BREEAM, launched in the UK in 1990, uses a lifecycle-based evaluation of energy, biodiversity, and water use, with certified projects achieving up to 25% carbon reductions [2,3]. LEED, developed by the U.S. Green Building Council in 1998, emphasizes carbon reduction and energy efficiency, showing energy savings up to 30% over non-certified buildings [2]. WELL, launched in 2014 by the International WELL Building Institute (IWBI), prioritizes health and well-being in the built environment. It evaluates indoor air quality, thermal comfort, and lighting but has been critiqued for a weaker focus on energy and embodied carbon [4,5].

Despite their strengths, all three systems face a persistent performance gap—the difference between design expectations and post-occupancy results. Discrepancies of 15–30% in energy use are common. Case studies show that operational performance depends more on ongoing monitoring and adaptive management than certification level. For example, Seattle’s Bullitt Center (LEED Platinum) achieved a low energy use intensity (16 kBtu/ft<sup>2</sup>/year) via real-time monitoring [6], while the LEED Gold David L. Lawrence Convention Center underperformed due to operational inefficiencies [7]. Similarly, the BREEAM-certified Bloomberg HQ in London achieved 30% lower energy use than peer buildings, and WELL-certified offices like Mirvac HQ and Delos HQ use real-time monitoring to optimize IEQ [8–10].

**Research Gap:** While comparative studies exist, most focus on design-stage criteria or isolated systems and neglect real-world outcomes or post-occupancy performance. Few integrate POE data or align certification performance with Sustainable Development Goals (SDGs). Moreover, no previous studies benchmark LEED, BREEAM, and WELL together across all three TBL pillars—environmental, social, and economic—using validated, operational data.

This study addresses these gaps by providing a comprehensive comparison of LEED, BREEAM, and WELL certifications through the Triple Bottom Line (TBL) framework. It offers three distinct contributions: (1) integration of POE data in evaluating certification outcomes, (2) alignment of results with SDGs 3 (health), 7 (clean energy), and 13 (climate action), and (3) application of a structured TBL benchmarking model.

The analysis uses a structured secondary review of the peer-reviewed literature, case studies, and POE data (2014–2024) to examine how well each certification aligns with long-term sustainability and operational performance targets.

By identifying strengths and systemic limitations, the study evaluates the real-world impact of LEED, BREEAM, and WELL across sustainability dimensions.

#### **Objectives:**

Evaluate how each certification system addresses environmental, social, and economic criteria via the TBL framework.

Compare performance in energy efficiency, carbon reduction, occupant well-being, and cost-effectiveness.

Investigate the causes of performance gaps between design and occupancy.

Identify best practices and limitations through real-world cases.

Propose recommendations to improve responsiveness and impact of certification systems.

#### **Research Questions:**

RQ1: How do LEED, BREEAM, and WELL differ in addressing environmental, social, and economic sustainability criteria under the TBL framework?

RQ2: What are the magnitudes and causes of the performance gap between certified design targets and actual post-occupancy outcomes?

RQ3: Which certification mechanisms most effectively align with post-occupancy performance data and SDG indicators?

## 2. Materials and Methods

### 2.1. Research Approach

This study conducted a structured secondary data analysis drawing from institutional certification platforms (e.g., USGBC, BRE, IWBI), peer-reviewed databases (e.g., Scopus, ScienceDirect), and public post-occupancy evaluation (POE) repositories. The inclusion criteria focused on (1) projects with LEED, BREEAM, or WELL certification; (2) availability of both design-stage targets and post-occupancy performance data; and (3) publication within the last 10 years (2014–2024). Excluded were projects lacking post-certification verification, single-family dwellings, or studies without standardized performance metrics. To ensure data validity, all sources were cross-referenced with third-party evaluations or peer-reviewed literature. Where discrepancies occurred, priority was given to independently verified outcomes over promotional claims [11–13].

Rather than relying on primary data, the research is theory-driven and utilizes official certification documentation, academic studies, and real-world case analyses to investigate each system's contributions to sustainable development and their alignment with the United Nations Sustainable Development Goals (SDGs) [14,15].

### 2.2. Data Sources

The dataset analyzed in this study includes approximately 50 documents, categorized into the following:

- Institutional materials from USGBC, BRE, and IWBI (e.g., rating manuals, case databases, technical briefs);
- Peer-reviewed journal articles, primarily retrieved via Scopus and ScienceDirect, focusing on certification outcomes and POE findings;
- Independent POE studies and third-party evaluation reports, including field data from academic and consultant sources.

Sources span from 2014 to 2024, with inclusion criteria requiring documentation of both design-phase projections and post-occupancy outcomes. Where multiple data points were available, preference was given to peer-reviewed or independently validated sources. The study synthesizes information from three validated source categories:

**Official Certification Manuals:** Technical guides and documentation from the U.S. Green Building Council (LEED), the Building Research Establishment (BREEAM), and the International WELL Building Institute (WELL) provided primary details about system structures, categories, and scoring criteria [16–18].

**Peer-Reviewed Literature:** Academic publications were selected for their focus on green building certification impacts, post-occupancy evaluation (POE), energy and lifecycle performance, and occupant well-being [12,13,19–21].

**Industry Case Studies:** Notable certified buildings—such as the Bullitt Center (LEED Platinum), Bloomberg HQ (BREEAM Outstanding), and Delos HQ (WELL Platinum)—were used to assess real-world operational outcomes and the persistent “performance gap” between design expectations and actual performance [6,8,10,22].

All data sources were evaluated for relevance, credibility, and contextual alignment with the TBL dimensions. The selection flow chart along with the quality assessment rubric for document selection by this study can be found in Appendix A.1 of this paper.

### 2.3. Analytical Framework

The evaluation process employed a two-phase analytical structure:

**System-Level Analysis:** Each certification system was examined independently to document its sustainability objectives, structural framework, and functional priorities. For instance, LEED emphasizes energy efficiency and site planning, but it also includes robust

biodiversity provisions—such as the ‘Protect or Restore Habitat’ credit in v4.1, which mandates native vegetation, habitat preservation, and pollinator support [12]. BREEAM similarly emphasizes lifecycle assessment and biodiversity, often through broader ecological and regional metrics [20].

**Comparative Evaluation via TBL:** The three systems were then assessed in parallel using the TBL framework as follows:

- **Environmental:** Metrics related to energy efficiency, water use, carbon emissions, and resource conservation [12,20];
- **Social:** Occupant well-being, indoor environmental quality (IEQ), equity, and mental health [13,21];
- **Economic:** Lifecycle cost efficiency, operational performance, and long-term resilience [19,22].

Each criterion was further substantiated by data from post-occupancy evaluations (POEs) where available.

#### *2.4. Inclusion and Exclusion Criteria*

The analysis focused exclusively on LEED, BREEAM, and WELL due to their international prominence, contrasting orientations, and established application across multiple geographic contexts [14,16]. While systems such as Green Star (Australia) and GSAS (Qatar/Gulf states) are recognized frameworks, they were excluded from this analysis due to the following two main factors:

(1) **Geographical focus:** This study concentrates on certification systems that are widely implemented across North America and Europe, where the majority of comparable post-occupancy data are available.

(2) **Data accessibility and compatibility:** Green Star and GSAS lacked sufficient peer-reviewed or independently verified post-occupancy evaluation (POE) data in the public domain that met our inclusion criteria. Their methodological structures and performance reporting frameworks also differ substantially, which would complicate cross-system benchmarking within the Triple Bottom Line (TBL) comparison.

#### *2.5. Study Limitations*

A primary limitation of this study lies in its reliance on secondary data, which may include inconsistent methodologies or limited access to building-specific POE results. Furthermore, the availability of real-time operational data for certified buildings is uneven, particularly across different regions and building typologies [19,22]. Nevertheless, this triangulated approach offers a structured basis for comparing certification system performance within a global sustainability framework.

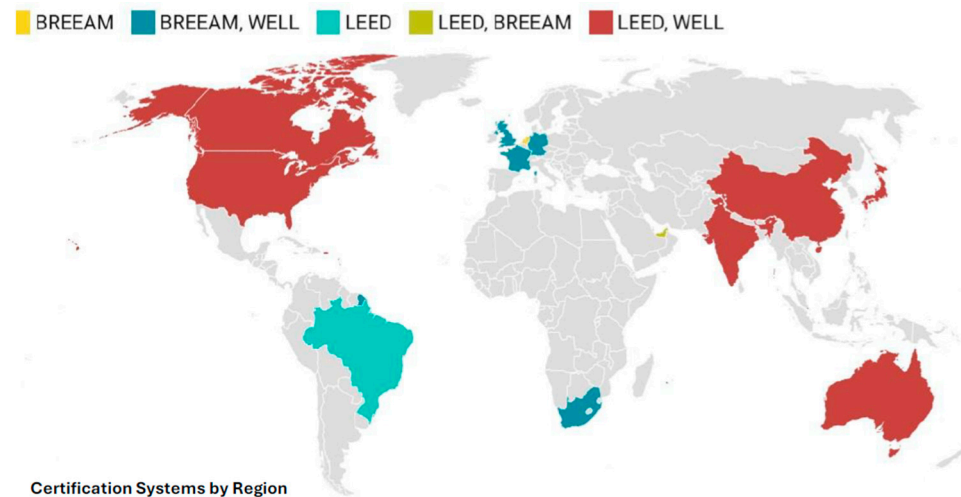
### **3. Benchmark and Comparative Analysis**

Green building certification systems have become foundational to global sustainability frameworks, providing the architecture, real estate, and infrastructure sectors with measurable, policy-aligned tools to reduce carbon, optimize resource use, and enhance occupant well-being. Among the most globally recognized frameworks—LEED, BREEAM, and WELL—each embodies a distinct approach to evaluating sustainability across environmental, social, and economic dimensions. This chapter presents a consolidated benchmark analysis, unifying findings from the literature review, technical manuals, post-occupancy evaluations, and regional policy studies, with a comparative lens informed by the Triple Bottom Line (TBL) framework.

### 3.1. Global Deployment and Regional Adaptation

LEED, developed by the U.S. Green Building Council, has achieved global reach with over 180 countries applying its framework, most strongly represented in the United States, United Kingdom, India, China, and the UAE; studies show that this increased knowledge and awareness affect the real estate market [16,23]. BREEAM, led by the UK's BRE, maintains strong application in Europe and has expanded through region-specific adaptations such as BREEAM-NL and BREEAM-Gulf [17]. WELL, though newer (launched in 2014), has seen accelerated uptake post-pandemic due to its occupant-centric health metrics, particularly in urban corporate sectors in North America, Australia, and Southeast Asia [18].

Figure 1 shows LEED-WELL dominance in North America, East Asia, and Oceania; BREEAM prevalence in Western Europe; and hybrid implementation in South Africa and Brazil. All the information has been collected from the certification systems' official websites and is presented in Table 1. The increasing convergence in regional systems is driven by climate-specific needs and the desire to align with national energy action plans (e.g., India's ECBC and the UK's Future Homes Standard) [24–27].



**Figure 1.** Global deployment of green building certification systems. This map illustrates the geographic distribution of LEED, BREEAM, and WELL-certified projects based on data from their respective official certification databases. Only countries with more than 50 certified buildings are displayed. Countries with smaller certification footprints are excluded for clarity.

**Table 1.** Sustainability certification-registered projects by country. The table shows the distribution of registered projects under LEED, BREEAM, and WELL certification systems across different countries.

| Country        | LEED    | BREEAM | WELL  | Total   |
|----------------|---------|--------|-------|---------|
| United States  | 35,000+ | 0      | 2000+ | 37,000+ |
| United Kingdom | 0       | 9000+  | 300+  | 9300+   |
| China          | 6000+   | 0      | 1000+ | 7000+   |
| India          | 5000+   | 0      | 300+  | 5300+   |
| UAE            | 1500+   | 100+   | 0     | 1600+   |
| Singapore      | 1500+   | 0      | 200+  | 1700+   |
| Germany        | 0       | 1000+  | 100+  | 1100+   |
| South Africa   | 0       | 500+   | 50+   | 550+    |

### 3.2. Comparative Evaluation of Certification Priorities

Each certification system exhibits internal prioritization of sustainability objectives. Each certification system emphasizes sustainability objectives through internally weighted criteria, as shown in their technical manuals and public guidance documentation. Table 2

below summarizes the comparative emphasis on core categories based on official scoring rubrics from LEED v4.1, BREEAM, and WELL [16–18]. These values were normalized to enable cross-system comparison.

**Table 2.** Comparison of certification criteria weighting across LEED, BREEAM, and WELL. Note: To enable cross-system comparison, category scores were normalized based on percentage contribution to each certification’s overall total. For example, a criterion worth 10 points out of 100 in LEED was treated as 10%, equivalent to a 10% weight in BREEAM or WELL. Where point systems were not directly comparable, alignment was based on thematic presence and relative weighting in the official documentation.

| Certification Criteria                                            | LEED (%) | BREEAM (%) | WELL (%) |
|-------------------------------------------------------------------|----------|------------|----------|
| Energy Efficiency                                                 | 33       | 25         | 10       |
| Indoor Environmental Quality (IEQ)                                | 15       | 15         | 30       |
| Water Efficiency                                                  | 10       | 6          | 5        |
| Materials and Resources                                           | 13       | 13         | 5        |
| Innovation/Design Process                                         | 6        | 10         | 10       |
| Health and Well-Being                                             | 5        | 8          | 30       |
| Location/Transport/Community                                      | 12       | 13         | 5        |
| Lifecycle Cost/Assessment                                         | 6        | 10         | 5        |
| Environmental Criteria Categories<br>(Energy + Water + Materials) | 56       | 44         | 20       |

Sources confirm that LEED places the greatest emphasis on operational energy savings [28], while BREEAM offers a more holistic lifecycle approach, incorporating embodied carbon and regional biodiversity [29]. WELL repositions the occupant as the central sustainability subject, anchoring its metrics around health outcomes, IEQ, and well-being [21,30].

A precise comparative evaluation of LEED, BREEAM, and WELL requires not only a surface comparison of credit weightings but also a deep examination of their criteria taxonomy, methodological constructs, and evaluation logics. Each certification system operationalizes sustainability differently across thematic domains, resulting in both epistemological divergence—differences in how sustainability is conceptualized—and functional asymmetries—differences in how performance is measured and verified. Table 3 summarizes the core mechanics of each system, including how points are awarded (basis of scoring), how compliance is verified, and the available certification levels. This comparative framing helps illustrate how methodological design affects the credibility, transparency, and replicability of sustainability outcomes across the three systems.

**Table 3.** Structural overview of certification systems. Note: Outlines the structural composition of each certification framework, including the basics of scoring, verification and certification levels.

| System | Basis of Scoring                                             | Verification                            | Certification Levels                          |
|--------|--------------------------------------------------------------|-----------------------------------------|-----------------------------------------------|
| LEED   | Point-based, prescriptive performance categories             | Independent third-party review via GBCI | Certified, Silver, Gold, Platinum             |
| BREEAM | Weighted category scores normalized to a % score             | Licensed Assessors + QA checks          | Pass, Good, Very Good, Excellent, Outstanding |
| WELL   | Preconditions + optimizations based on performance standards | IWBI verification, sensors, surveys     | Silver, Gold, Platinum                        |

The criteria across the three systems cluster around seven major thematic areas. However, each certification system defines and implements these categories with varying levels of scope, depth, and stringency. Table 4 was developed through cross-analysis of certification manuals, project documentation, and post-occupancy evaluations from certified buildings. The table illustrates how each system addresses these themes—such as energy, water, or health—through distinct operational strategies and credit structures. This comparative mapping reveals areas of overlap as well as systemic gaps, particularly in domains like lifecycle integration and health-energy trade-offs. Tables 3 and 4 serve distinct but complementary functions. While Table 3 provides a structural comparison of certification systems (scoring logic, verification mechanisms, and certification tiers), Table 4 examines how each system addresses core sustainability domains. This allows for both macro-level (system structure) and micro-level (category emphasis) comparison.

**Table 4.** Thematic category mapping and emphases of each certification system.

| Domain                                                         | LEED                                            | BREEAM                                                                                 | WELL                                       |
|----------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Energy and Carbon</b>                                       | ASHRAE-based modeling, renewable energy credits | Whole-building simulation, LCA of energy                                               | Indirect; via comfort controls             |
| <b>Indoor Environmental Quality</b>                            | Low-emitting materials, ventilation, daylight   | Acoustic, thermal, IAQ, visual comfort                                                 | Dominant focus: air, light, sound, comfort |
| <b>Water Efficiency, Water Quality and Water Accessibility</b> | Fixture flow rates, reuse, metering             | Water quality tested, microbial safety; accessible handwashing and drinking facilities | Drinking water purity; access              |
| <b>Materials and Resources</b>                                 | Life-cycle impact (EPD), recycled content       | Sourcing, durability, toxicity                                                         | Limited to material toxicity               |
| <b>Health and Well-Being</b>                                   | Innovation credits for biophilia, ergonomics    | Credits on user health are less emphasized                                             | Central domain: 100+ optimization options  |
| <b>Management and Operations</b>                               | Commissioning, metering, building operations    | Mandatory management processes                                                         | Occupant behavior; HR policies             |
| <b>Post-Occupancy Evaluation</b>                               | Optional via M&V credit (IPMVP-based)           | Operational performance review (soft POE)                                              | Mandatory sensors and occupant feedback    |

LEED adopts a performance-anchored, engineering-led model centered around operational energy optimization and material impact, but its IEQ provisions are largely prescriptive and not dynamically verified post-occupancy. Critics have described it as “quantitative but weak in longitudinal tracking” [28].

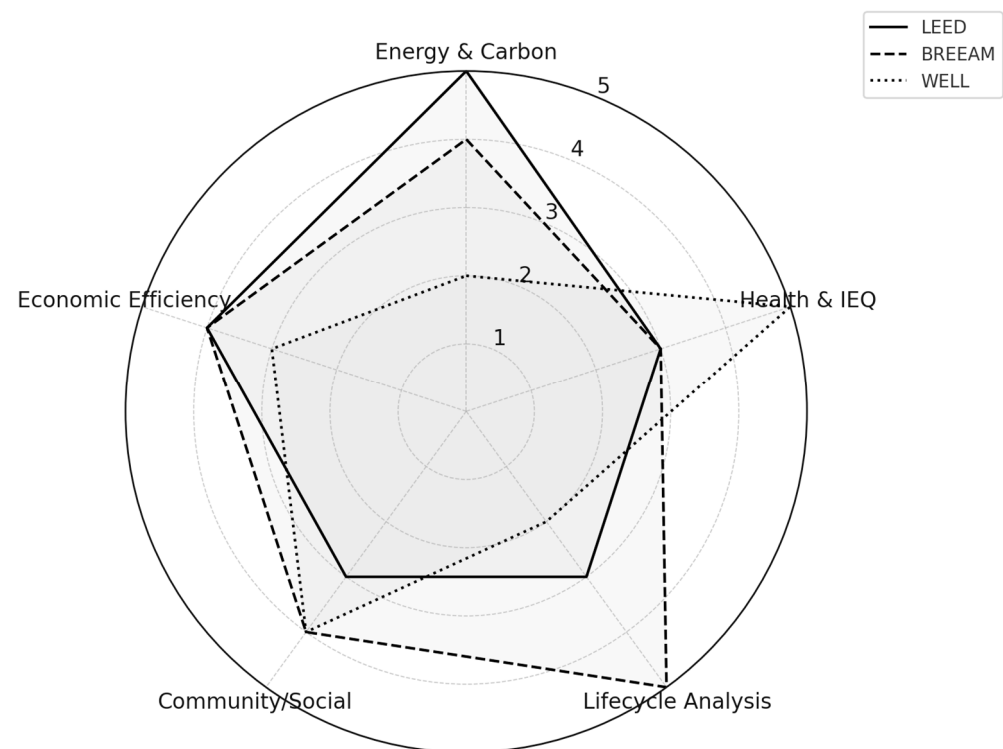
BREEAM introduces a hybrid model blending LCA with regional weighting schemes, especially strong in Europe. Its ecological criteria (e.g., biodiversity, flood resilience) extend beyond building-centric scopes. However, its complex documentation protocols can hinder adoption outside the UK [29].

WELL reframes sustainability as a health science issue, grounding its approach in evidence-based medicine, human biology, and biophilic design theory. While its feed-

back mechanisms (e.g., SAMBA sensors, user surveys) enhance IEQ credibility, it lacks substantive modeling of energy or carbon intensity [30].

### 3.3. Triple Bottom Line Benchmarking

A radar chart comparison (Figure 2) further illustrates the differentiated strengths of each certification scheme.



| Score | Interpretation                                                                            |
|-------|-------------------------------------------------------------------------------------------|
| 0     | Not addressed by the certification system                                                 |
| 1     | Minimally addressed, no mandatory credit                                                  |
| 2     | Addressed as an optional or weak credit                                                   |
| 3     | Moderately addressed with standard compliance                                             |
| 4     | Strongly addressed, integrated into the core system                                       |
| 5     | Central focus of system with detailed metrics, enforcement, and post-occupancy validation |

**Figure 2.** Radar chart comparison of LEED, BREEAM, and WELL across core TBL metrics. Note: Scores were derived from core criteria in each system, enabling cross-framework comparison.

The radar chart presented in Figure 2 offers a visual synthesis of how LEED, BREEAM, and WELL certification systems perform across five core sustainability metrics derived from the Triple Bottom Line framework: Energy and Carbon, Health and Indoor Environmental Quality (IEQ), Lifecycle Analysis, Community/Social Integration, and Economic Efficiency.

Each axis on the radar represents a normalized score (0–5), constructed from an aggregate of credit weightings, published performance data, and peer-reviewed case study benchmarks [28–33].

LEED demonstrates the highest performance in the Energy and Carbon category due to its strict requirements for energy modeling, site optimization, and renewable integration. Projects such as the Bullitt Center highlight LEED’s effectiveness in achieving ultra-low energy use intensities (EUIs) when paired with performance tracking [22,28,34,35]. It also shows solid scores in Economic Efficiency, driven by energy cost savings and demand-side

management programs. However, LEED is relatively weaker in Lifecycle Analysis and Health and IEQ, where its metrics are optional or marginally weighted [31].

BREEAM ranks highest in Lifecycle Analysis, emphasizing full lifecycle carbon accounting, embodied emissions, and biodiversity preservation [29]. Its system rewards the use of locally sourced materials, passive systems, and long-term design resilience. It also fares well in Community/Social and Economic Efficiency, particularly in European contexts where policies like the EU Taxonomy influence building codes [32]. Its performance in Health and IEQ, however, while included, is not as extensively developed as in WELL.

WELL dominates the Health and IEQ axis, with a high concentration of metrics focused on thermal comfort, acoustic quality, daylighting, mental health, and air purity [30,33]. Unlike LEED or BREEAM, WELL includes sensor-based verification and user experience surveys as part of its validation process. It also performs well in Community/Social, particularly through inclusive measures and behavioral wellness programming. Nevertheless, WELL underperforms in Energy and Carbon and Lifecycle Analysis due to limited environmental depth and the absence of requirements for embodied energy or operational carbon tracking.

Overall, the radar visualization underscores the complementary nature of these systems: while none fully satisfies all sustainability dimensions, their strengths are distinct and, when combined, can provide comprehensive coverage of TBL criteria. This supports growing advocacy for dual certification approaches, such as LEED+WELL or BREEAM+WELL, in delivering holistic sustainability outcomes.

#### 3.4. Post-Occupancy Evaluation and Performance Gaps

Despite rigorous pre-certification modeling and sustainability forecasting in green building certification systems (GBCs), a persistent “performance gap” remains between predicted and actual building performance [34]. This gap undermines the credibility and effectiveness of even the most lauded rating systems, such as LEED, BREEAM, and WELL, highlighting the need for more robust post-occupancy evaluation (POE) practices [30,36].

POE is a structured methodology for assessing building performance under actual occupancy conditions, capturing metrics such as energy consumption, occupant satisfaction, indoor environmental quality (IEQ), and system functionality [30]. Unlike static certification checklists, POE offers dynamic feedback for operational adjustments.

LEED’s POE data often reveal mismatches in HVAC and lighting efficiency due to occupant behavior. BREEAM’s POEs emphasize lifecycle and water performance but often omit comfort data. WELL focuses more on health-related metrics like air quality and acoustics. These patterns inform the comparative analysis that follows.

##### 3.4.1. Key Case-Based Insights on POE Effectiveness

Table 5 presents POE results from four certified buildings. They illustrate how POE validates or challenges design-stage assumptions.

These examples show that certification alone does not ensure post-occupancy success. Studies show 30–50% of LEED buildings underperform [35,36], often due to behavioral, commissioning [30], or maintenance gaps.

In WELL-certified environments, high comfort standards may unintentionally increase energy demand, showing a trade-off between health and environmental priorities [5,37]. In contrast, buildings like the Bullitt Center demonstrate best practices by integrating POE into ongoing management, often exceeding design targets via smart metering and feedback systems [38–43].

**Table 5.** Post-occupancy evaluation (POE) outcomes in certified buildings.

| Building                                   | Certification      | POE Impact                                                                                                        | Key Metrics                              | Ref. |
|--------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------|------|
| <b>Bullitt Center, Seattle</b>             | LEED Platinum      | Real-time POE and energy metering allowed an EUI of 16 kBtu/ft <sup>2</sup> /year, surpassing modeled performance | Energy metering, HVAC tuning             | [35] |
| <b>David L. Lawrence Convention Center</b> | LEED Gold          | Underperformed due to HVAC errors and staff training deficits; POE revealed systemic gaps                         | HVAC inefficiency, staff behavior        | [36] |
| <b>Bloomberg HQ, London</b>                | BREEAM Outstanding | POE-driven recalibration of lighting and ventilation saved 35% over predicted energy usage                        | Automated controls, daylight integration | [22] |
| <b>Delos HQ, New York City</b>             | WELL Platinum      | Comfort sensors improved IEQ but raised HVAC energy usage, indicating a comfort-energy trade-off                  | IEQ sensors, energy-thermal balancing    | [10] |

### 3.4.2. Implications for Certification Frameworks

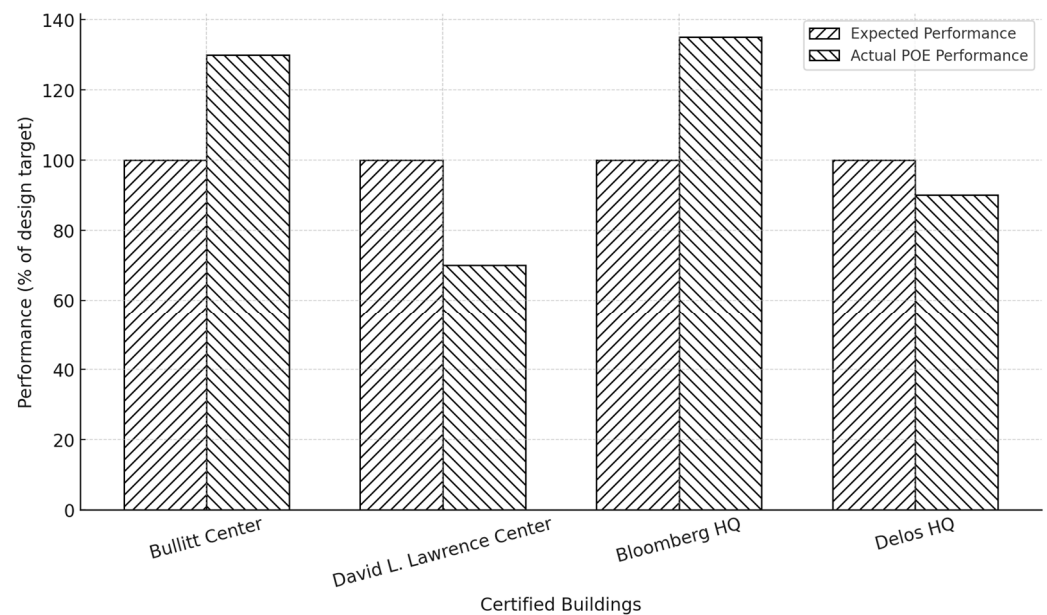
The consistent emergence of the performance gap across leading rating systems exposes a fundamental weakness in static, checklist-driven assessment models. As such, scholars and industry experts advocate the integration of POE into certification recertification loops, enabling the following:

- ✓ Data-driven corrections during operation;
- ✓ Verification of real-time energy and IEQ metrics;
- ✓ Stakeholder-responsive facility management;
- ✓ Iterative design improvements across building portfolios [44–46].

To enhance credibility and sustainability alignment, certifications must evolve toward a hybrid model—combining design compliance with longitudinal operational analytics.

As shown in Figure 3, POE outcomes vary widely across certified projects. The Bullitt Center outperformed its energy targets by ~30% due to integrated monitoring [35]. The David L. Lawrence Center underperformed by ~30%, citing HVAC and training failures [36]. Bloomberg HQ, BREEAM Outstanding, saved 35% with POE-driven recalibration [29]. Delos HQ (WELL Platinum) underperformed by ~10%, due to a trade-off between enhanced occupant comfort and energy use; real-time environmental controls improved thermal and acoustic conditions but inadvertently increased HVAC energy intensity [33]. A comparative and summative table between case studies can be found in Appendix A.2.

These results underscore the heterogeneous nature of certification outcomes, validating that high design scores alone are insufficient predictors of real-world sustainability performance. Instead, they emphasize the imperative role of operational transparency, continuous metering, and feedback-driven building management in fulfilling the holistic goals of green building certification systems.



**Figure 3.** Post-occupancy evaluation (POE) performance versus design expectations across selected LEED, BREEAM, and WELL-certified buildings. Performance above 100% indicates outperformance; below 100% indicates a shortfall relative to predicted performance targets. *Note: The presented values reflect approximate average post-occupancy outcomes drawn from available case studies. Due to inconsistent reporting of variance or sample sizes across sources, error bars or confidence intervals could not be reliably included. As such, the chart conveys general performance trends rather than statistically precise estimates.*

### 3.5. SDG Alignment and Sustainability Leadership

The systems also vary in how well they align with the United Nations Sustainable Development Goals (SDGs). The alignment ratings in Table 6 were derived through a structured review of each certification system’s documentation and performance reports, assessing how explicitly and extensively their criteria address the targets of each SDG. “Strong” alignment indicates direct coverage of core SDG themes with substantial weighting or mandatory criteria (e.g., WELL’s emphasis on SDG 3: Health and Well-Being). “Medium” reflects partial or indirect inclusion, and “Weak” indicates minimal relevance or optional treatment. These assessments were further supported by post-occupancy evaluations and literature comparing real-world project outcomes against sustainability targets.

**Table 6.** SDG alignment of certification systems (*white circles: weak & medium alignment, green circles: strong alignment*).

| SDG                            | LEED   | BREEAM | WELL   |
|--------------------------------|--------|--------|--------|
| SDG 3—Good Health              | Medium | Medium | Strong |
| SDG 7—Clean Energy             | Strong | Strong | Weak   |
| SDG 11—Sustainable Cities      | Strong | Strong | Strong |
| SDG 12—Responsible Consumption | Medium | Strong | Weak   |
| SDG 13—Climate Action          | Strong | Strong | Medium |

As shown in Table 6, each system exhibits unique alignments with specific SDG targets, shaped by its methodological priorities, indicator selection, and certification culture. More specifically, WELL dominates in SDG 3 (health), while BREEAM is aligned with SDGs 12 and 13 through its resource and carbon tracking systems. LEED strongly targets SDG 7 and 13 via renewable energy and GHG reduction requirements [14,28–30].

The alignment of LEED, BREEAM, and WELL with the United Nations Sustainable Development Goals (SDGs) provides an essential lens to evaluate their broader societal and ecological contributions beyond operational performance. However, due to the absence of standardized SDG measurement protocols across certifications, alignment ratings are based on qualitative interpretation.

### **SDG 3—Good Health and Well-Being**

The WELL Building Standard demonstrates a robust commitment to SDG 3, prioritizing occupant health outcomes through its emphasis on indoor air quality, thermal comfort, lighting, acoustic performance, mental health, and fitness-supportive infrastructure [5,9,10,28,33]. The system's reliance on post-occupancy surveys and sensor-based IEQ tracking (e.g., SAMBA at Mirvac HQ [9]) ensures that WELL-certified spaces continuously support physiological and psychological well-being. Unlike LEED and BREEAM, WELL's structure explicitly integrates biophilic design, non-toxic materials, and mental restoration elements, which are essential under WHO's Healthy Buildings framework [42].

### **SDG 6—Clean Water and Sanitation**

BREEAM and LEED align moderately with SDG 6 by incorporating water efficiency metrics, such as rainwater harvesting, low-flow fixtures, and greywater reuse [24,31]. BREEAM's regional adaptations (e.g., BREEAM Gulf) further strengthen this alignment by addressing water scarcity through context-specific water benchmarks [47]. WELL, while considering potable water quality for user health, it lacks broader water conservation strategies, limiting its impact on systemic water sustainability [33].

### **SDG 7—Affordable and Clean Energy**

LEED is the most aligned with SDG 7, especially through its Energy and Atmosphere category, which rewards projects that integrate on-site renewables, passive design strategies, energy modeling, and smart metering [24,28,31]. The Bullitt Center case demonstrates LEED's potential in operationalizing energy autonomy, where performance-based credits incentivize design optimization across energy use intensity (EUI) and net-zero readiness [35]. BREEAM similarly integrates renewable targets, though its scoring is more procedural and lifecycle-focused [29,47].

### **SDG 11—Sustainable Cities and Communities**

All three systems contribute to SDG 11, though in differing ways. LEED and BREEAM support urban sustainability through categories on site location, transportation, resilience planning, and community connectivity [24,29]. BREEAM's urban planning modules (e.g., "Land Use and Ecology") provide tools to reduce urban heat island effects, promote biodiversity corridors, and improve walkability, particularly in master planning projects [32,34]. WELL, on the other hand, engages with this SDG through the lens of user inclusivity, accessibility, and mental wellness in dense urban settings, enhancing social sustainability dimensions often neglected in traditional environmental tools [30,33,42].

### **SDG 12—Responsible Consumption and Production**

BREEAM stands out in its comprehensive integration of lifecycle assessment (LCA), embodied carbon, and circular economy principles, making it the most relevant certification under SDG 12 [32,34]. Its Material Efficiency and Waste categories reward local sourcing, adaptive reuse, and dematerialization—aligning with EU goals for resource neutrality. LEED addresses material sustainability as well but lacks the granularity of BREEAM's end-of-life and supply chain impact scoring [24,26]. WELL has limited alignment with SDG 12, focusing more on human-centric aspects rather than resource systemicity [33].

### **SDG 13—Climate Action**

Both LEED and BREEAM are directly aligned with SDG 13 by integrating greenhouse gas (GHG) assessments, carbon mitigation strategies, and climate resilience planning [24,29,32]. LEED emphasizes operational carbon, including tools for energy simulation

and emissions modeling (e.g., ASHRAE 90.1 compliance), while BREEAM expands its scope to embodied carbon and material sourcing emissions. Projects certified under these systems often contribute directly to national climate targets or green finance initiatives. WELL, despite offering health-oriented climate adaptation, it lacks direct carbon measurement or energy reduction obligations, thereby making a limited contribution to this goal [33].

### *3.6. Limitations, Regionalization, and Systemic Divergences*

Despite growing global uptake, green building certification systems (GBCSs) such as LEED, BREEAM, and WELL continue to exhibit significant adaptation challenges when deployed across varied geographical, climatic, and regulatory contexts. These limitations affect both the technical applicability of sustainability metrics and the cultural alignment of health, comfort, and social well-being parameters.

#### **Limitations in Geographic Portability and Climatic Sensitivity**

LEED, originating in the U.S., remains predominantly structured around North American codes, energy baselines (e.g., ASHRAE), and climate assumptions, creating friction when applied in European, African, and Middle Eastern regions. In arid or tropical climates, for example, LEED's emphasis on HVAC efficiency and solar shading may not align with local building norms or passive design traditions, often leading to low local relevance and high cost of compliance.

BREEAM, by contrast, adopts a regional modular model, with variants like BREEAM International, BREEAM Gulf, and BREEAM NOR, designed to recalibrate baseline metrics such as water availability, solar exposure, and indigenous biodiversity. These schemes demonstrate superior contextual flexibility, for example, using native vegetation baselines and local thermal comfort indices in arid zones. However, critics argue that BREEAM's documentation-heavy approach and complex scoring can deter widespread adoption in emerging markets.

WELL faces its own regional limitations, particularly in hot, humid, or high-heat regions. In countries like the UAE, Saudi Arabia, or India, maintaining WELL-compliant levels of thermal comfort, indoor air quality, and natural ventilation can paradoxically result in higher HVAC-related energy use, undercutting sustainability gains. This highlights a core conflict between comfort-centric design and carbon minimization, a trade-off not yet reconciled in WELL's standard structure.

#### **Systemic Divergences: Integrative Challenges**

From a systems integration standpoint, the fragmentation of sustainability dimensions across LEED (environmental), BREEAM (lifecycle and governance), and WELL (health and wellness) creates functional silos. As urban environments become more complex, research increasingly calls for hybridized frameworks that blend LEED's energy and water rigor with BREEAM's lifecycle realism and WELL's occupant-centered strategies. For example, combining POE-backed operational performance from LEED with WELL's real-time environmental sensor feedback could create continuous verification loops, reducing the performance gap.

The need for this integration is also reflected in growing climate-health convergence frameworks, where carbon reduction, heat resilience, and social equity intersect. However, as yet, no certification program offers a truly cross-cutting, SDG-optimized solution adaptable across global bioclimatic and socioeconomic zones.

### *3.7. Conclusions of the Benchmark Comparative Analysis*

This benchmark and comparative analysis reveal that no single certification system provides holistic coverage across all sustainability domains. Each excels in the following distinct aspects:

- LEED leads in operational energy efficiency, emissions tracking, and material optimization, aligning most strongly with SDGs 7 and 13.
- BREEAM outperforms in lifecycle analysis, circular economy integration, and biodiversity, making it a natural fit for SDGs 12 and 15.
- WELL remains unmatched in advancing occupant well-being, mental health, and indoor environmental quality, making it the leading tool for realizing SDG 3.

Nevertheless, performance gaps persist, and regional incompatibility reduces efficacy without targeted localization. The lack of post-occupancy adaptation mechanisms and dynamic feedback loops continues to limit all three systems' potential in climate-responsive design. To move forward, the field must embrace multi-criteria certification pathways, where credits are not merely additive but interoperable across tools. Future green building certifications should be incorporated as follows:

- Region-specific baselines rooted in local climatic, cultural, and ecological parameters.
- Cross-certified POE integration to track actual use vs. design targets.
- Unified indicators aligned with SDG metrics, to facilitate global benchmarking and ESG reporting.

Ultimately, a next-generation certification ecosystem should be modular, adaptive, and performance-verified, blending environmental resilience with social well-being and economic efficiency in ways today's siloed frameworks still fall short of achieving.

#### 4. Results and Discussion

The comparative evaluation of LEED, BREEAM, and WELL across global benchmark cases reveals distinct orientations in certification methodology and thematic emphasis. These systems operate under structurally divergent paradigms: LEED is primarily performance-anchored in energy and materials; BREEAM adopts a lifecycle and ecological footprinting lens; and WELL redefines building sustainability through the health and psychological well-being of occupants.

The case studies confirm that LEED-certified buildings consistently demonstrate superior outcomes in energy modeling and carbon intensity reduction, attributed to rigorous baseline modeling aligned with ASHRAE standards. BREEAM-certified projects, such as Bloomberg HQ, reveal deeper engagement with environmental management protocols, biodiversity considerations, and post-construction operational guidance [47]. WELL buildings, on the other hand, exhibit the strongest performance in indoor environmental quality and user-centric design, yet often lack the embedded carbon tracking or LCA stringency found in LEED or BREEAM [48].

Despite advancements in certification frameworks, performance gaps persist due to several systemic factors. First, systems like LEED emphasize design-phase modeling over real-world outcomes, contributing to discrepancies between predicted and actual performance—as seen in the 30% energy shortfall at the Lawrence Center [36] and supported by simulation critiques in prior work [7]. Second, occupant behavior introduces variability, with inefficiencies often stemming from poor training or manual overrides of automated systems [49,50]. Third, post-certification performance declines when commissioning and maintenance protocols are insufficient, as noted in both this study and supporting literature [31]. WELL-certified buildings such as Delos HQ [10, further highlight a comfort–energy trade-off, where enhanced thermal and acoustic control leads to increased HVAC loads—a pattern confirmed by Allen et al. [30] and Ildırı et al. [5]. Finally, post-occupancy evaluation (POE) remains inconsistently integrated: while WELL includes POE in its framework, LEED and BREEAM treat it as optional, weakening accountability for real-world performance [43]. These findings are summarized in Table 7, while highlighting

the need for certification systems to evolve from static design compliance toward dynamic, feedback-driven models that prioritize actual building outcomes.

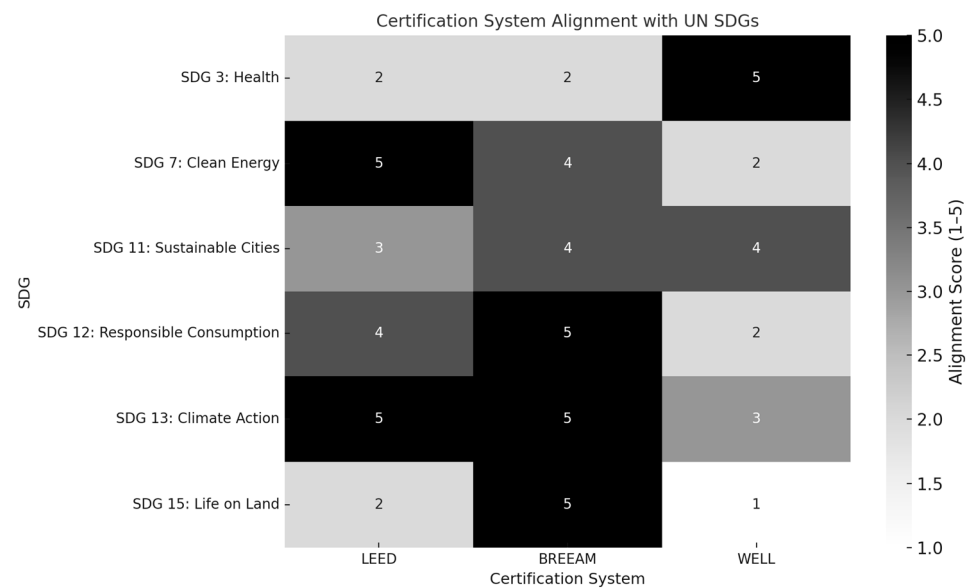
**Table 7.** Summary of identified performance gaps and supporting evidence from previous studies.

| Root Cause of Performance Gap                                       | Study Findings                                                                          | Supporting Literature | Notes                                                                  |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------|
| <b>Design-phase modeling is prioritized over actual performance</b> | LEED buildings (e.g., Lawrence Center) underperformed by 30% despite high certification | [7]                   | Predictive models are often too optimistic; limited post-cert tracking |
| <b>Occupant behavior variability</b>                                | HVAC inefficiencies linked to poor training and override in LEED Gold projects          | [50]                  | User actions strongly impact operational outcomes                      |
| <b>Lack of post-handover commissioning and maintenance</b>          | Systemic gaps in underperforming projects (e.g., Lawrence Center)                       | [31]                  | Commissioning gaps reduce system effectiveness                         |
| <b>Comfort-energy trade-off in WELL</b>                             | Delos HQ used more energy (+10%) to meet WELL comfort goals                             | [5,30]                | Comfort-driven systems raise energy intensity                          |
| <b>Weak integration of POE in recertification</b>                   | WELL includes POE mechanisms; LEED/BREEAM treats it as optional                         | [43]                  | POE is rarely required for certification renewal                       |

The radar plot comparison (Figure 4) illustrates these differences quantitatively. LEED scored highest on energy and materials; BREEAM led in lifecycle and management practices; WELL scored maximally in health and user feedback. The disparity in coverage suggests that these systems are not interchangeable. Rather, they act as complementary frameworks, each excelling in certain domains while leaving others underrepresented.

Further insights emerge when certification coverage is compared against selected Sustainable Development Goals (SDGs). The alignment matrix (Figure 4) confirms that no certification system addresses the SDGs holistically. LEED performs strongly on SDG 7 (Clean Energy) and SDG 12 (Responsible Consumption), while BREEAM shows alignment with SDG 13 (Climate Action) and SDG 15 (Life on Land), largely due to its ecological design incentives [49]. WELL, in contrast, aligns more directly with SDG 3 (Good Health and Well-Being) and SDG 11 (Sustainable Cities), supported by its focus on biophilic design, air quality, and post-occupancy surveys [50].

Post-occupancy evaluation (POE) emerges as a core divergence point. While WELL mandates POE through integrated feedback systems and biometric metrics, LEED and BREEAM consider it optional or limited to commissioning stages. The absence of longitudinal verification in LEED- and BREEAM-certified buildings raises concerns about performance decay and occupant dissatisfaction over time—a gap well documented in prior studies [51,52].



**Figure 4.** Certification system alignment with UN SDGs (1 = Low, 5 = High).

These findings directly address the original research questions posed in the introductory chapter. Certification systems are not functionally equivalent, and each prioritizes a different axis of sustainability. Their limitations, particularly in POE, SDG traceability, and social governance, undermine efforts toward comprehensive, measurable sustainability transitions in the built environment. Given these results, several recommendations are proposed below:

#### 1. Hybrid Certification Models

Projects should consider pursuing combined certifications (e.g., LEED + WELL or BREEAM + WELL) to leverage the strengths of both environmental performance and user-centered metrics. This hybridization strategy ensures greater TBL coverage without duplicative effort.

#### 2. Mandatory Post-Occupancy Evaluation

POE should become a structural requirement, not elective credit. Longitudinal data—collected through environmental sensors, occupant surveys, and real-time feedback—are essential to align operational outcomes with design intent [53].

#### 3. SDG-Based Credit Structuring

Certification bodies should integrate SDG mapping directly into credit systems. Projects could then report their SDG contributions through standardized, verifiable indicators that would enhance both policy transparency and stakeholder accountability [54].

#### 4. Regional Contextualization

Building on BREEAM's localized weightings, certification schemes should allow regional climate, policy, and cultural differences to inform credit valuation. A Mediterranean office should not be evaluated with the same criteria as a Scandinavian school [55].

#### 5. Feedback-Driven Recertification

Systems should adopt a cyclical recertification model that is contingent upon performance metrics, not time alone. Smart technology now enables this possibility on a scale and would allow more adaptive, responsive sustainability governance [56].

The implications of these recommendations extend to design teams, certifying bodies, policymakers, and occupants themselves. Buildings are no longer static infrastructure

but living systems that mediate environmental, social, and economic feedback. As such, their certification should reflect this dynamism in structure, logic, and methodology. In particular, this study's finding could have the following actionable recommendations:

- Design teams should incorporate adaptive systems with POE-readiness from the outset.
- Certification bodies should make POE mandatory for recertification and tie credits to verified outcomes.
- Policymakers should develop incentives for verified post-occupancy performance.
- Occupants themselves could be favorable to providing feedback loops via apps or dashboards and engaging in energy use education.

The data support the original research questions: certification systems vary significantly in focus, overlap partially with SDGs, and are often siloed in disciplinary scope. Their current design limits cross-pillar integration. For example, WELL achieves high satisfaction rates in thermal and visual comfort but does not impose strong requirements on embodied carbon or lifecycle durability. Conversely, LEED may deliver superior resource efficiency without systematically assessing user comfort over time.

Ultimately, the results underscore the need for a systemic redesign of certification logic. Cross-system interoperability, dynamic POE requirements, and SDG-traceable credit structures are necessary to support comprehensive sustainability outcomes in the built environment.

To aid stakeholders in selecting the most appropriate certification framework, Table 8 summarizes the contexts and project types where each system offers the greatest value. Overall, the case study analysis supports the conclusion that certification systems are most effective when aligned with project context and performance priorities. LEED demonstrates strong outcomes in energy-intensive urban office buildings due to its focus on operational efficiency and carbon mitigation. BREEAM's strength in lifecycle assessment and biodiversity planning makes it particularly suitable for public-sector and institutional projects in policy-driven contexts such as the UK and EU. Meanwhile, WELL certification delivers the greatest value in human-centered environments—including schools, clinics, and tech workplaces—where health, well-being, and indoor environmental quality (IEQ) are core design imperatives. These context-specific applications also correlate with lower post-occupancy performance gaps, underscoring the importance of selecting standards that match real-use conditions and stakeholder priorities.

However, while integrating post-occupancy evaluation (POE) and real-time performance tracking into certification frameworks is conceptually valuable, several practical barriers limit implementation. First, many buildings lack the infrastructure—such as sub-metering, environmental sensors, or digital commissioning tools—needed to capture reliable post-occupancy data. Second, data privacy concerns and inconsistent reporting standards across building types and regions complicate benchmarking and public disclosure. Third, responsibility for long-term building performance is often fragmented among design teams, contractors, facility managers, and occupants, making accountability difficult to enforce. Finally, there is a shortage of trained professionals with the interdisciplinary skills required to interpret POE data and translate findings into actionable improvements. These challenges highlight the need for supportive policies, education, and funding mechanisms to enable a meaningful shift from design-based certification to performance-driven accountability.

**Table 8.** Recommendations on when/where to apply each standard.

| Certification                                                                   | Strengths                                                                                           | Best-Suited Contexts                                                                                                                            |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LEED</b> (Leadership in Energy and Environmental Design)                     | Strong focus on operational energy reduction, carbon footprint mitigation, and materials selection. | High-performance commercial and institutional buildings; net-zero energy targets; global projects seeking international recognition.            |
| <b>BREEAM</b> (Building Research Establishment Environmental Assessment Method) | Emphasizes lifecycle carbon accounting, biodiversity protection, and contextual adaptability.       | Projects in Europe or regions with mature LCA tools; public sector developments; urban regeneration with ecological integration.                |
| <b>WELL</b> (WELL Building Standard)                                            | Prioritizes occupant health, indoor environmental quality (IEQ), and post-pandemic resilience.      | Office renovations, healthcare facilities, and educational buildings aiming for health-centered performance; post-COVID-19 workplace retrofits. |

## 5. Conclusions

This study presents a comprehensive comparative assessment of three dominant green building certification systems—LEED, BREEAM, and WELL—evaluating their structure, sustainability depth, health-centered features, and real-world performance outcomes. Building upon a triangulated benchmarking framework developed in the research, the analysis integrates key criteria across energy performance, indoor environmental quality (IEQ), lifecycle assessment (LCA), human well-being, and alignment with the Sustainable Development Goals (SDGs).

The benchmarking model revealed the following:

- LEED offers broad international applicability with strong energy modeling protocols. However, it remains partially prescriptive and underperforms in adaptive post-occupancy feedback systems [57,58].
- BREEAM is robust in lifecycle criteria and biodiversity integration, aligning closely with EU environmental policies. Nevertheless, its documentation-intensive nature and region-specific tools reduce its adoption in global markets [59].
- WELL excels in occupant health, biophilic design, and active sensing strategies, yet its disconnection from carbon metrics and energy efficiency standards limits its comprehensive sustainability scope [60,61].
- Across all certifications, the lack of longitudinal post-occupancy evaluation (POE) remains a recurring performance gap, affecting reliability in real-world impact assessments [62].

To move toward performance-driven and SDG-aligned certification, the study proposes the following four targeted recommendations:

- Bridge the energy–health divide by combining WELL’s sensor-based comfort metrics with LEED/BREEAM energy protocols [63].
- Promote open data benchmarking, enabling cross-project transparency in operational carbon, comfort, and wellness [64].
- Link certification credits explicitly to SDG indicators, especially SDGs 3, 7, 11, and 13, for traceable global impact [65].
- Integrate POE into certification renewal cycles, using real-time metrics to verify actual impact [66].

The research identifies several key directions for future academic and industry exploration. First, longitudinal studies are essential to track how certified buildings perform over time in energy consumption, IEQ, and occupant health, addressing gaps in post-certification accountability [67,68]. Second, applying machine learning to POE data could reveal latent correlations between design intentions and real outcomes through telemetry and occupant feedback systems [69]. Third, cross-jurisdictional policy analysis should explore how national codes (e.g., India’s ECBC, UK’s Future Homes Standard) align—or conflict—with international certification logic [70,71]. Finally, context-specific adaptations for the Global South are a critical avenue for equitable decarbonization, requiring flexible tools that acknowledge economic and climatic disparities [72–74].

In summary, sustainability certification systems have evolved from prescriptive checklists into strategic tools of global urban governance, but their success hinges on deeper integration of data, accountability, and long-term social equity. This study urges a transformation toward science-based benchmarking, human-centric metrics, and deeper alignment with the global SDG framework to ensure certifications do not only promise sustainability but deliver it.

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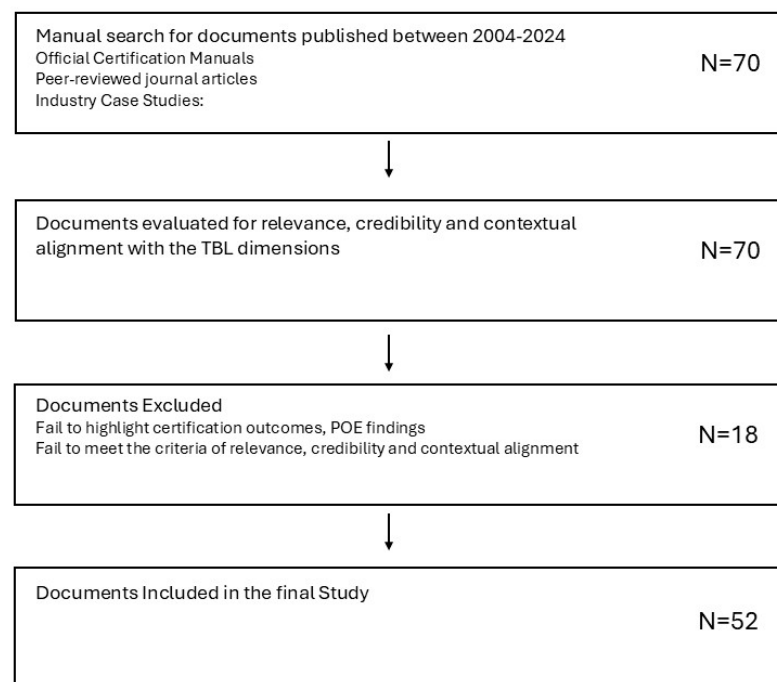
## Abbreviations

The following abbreviations are used in this manuscript:

|                 |                                                                           |
|-----------------|---------------------------------------------------------------------------|
| LEED            | Leadership in Energy and Environmental Design                             |
| BREEAM          | Building Research Establishment Environmental Assessment Method           |
| WELL            | WELL Building Standard                                                    |
| GBCSs           | Green Building Certification Systems                                      |
| TBL             | Triple Bottom Line                                                        |
| IEQ             | Indoor Environmental Quality                                              |
| LCA             | Life Cycle Assessment                                                     |
| POE             | Post-Occupancy Evaluation                                                 |
| SDG             | Sustainable Development Goal                                              |
| HVAC            | Heating, Ventilation, and Air Conditioning                                |
| ASHRAE          | American Society of Heating, Refrigerating and Air-Conditioning Engineers |
| IWBI            | International WELL Building Institute                                     |
| USGBC           | U.S. Green Building Council                                               |
| BRE             | Building Research Establishment                                           |
| EUI             | Energy Use Intensity                                                      |
| EPBD            | Energy Performance of Buildings Directive                                 |
| BIM             | Building Information Modeling                                             |
| M&V             | Measurement and Verification                                              |
| LCC             | Life Cycle Cost                                                           |
| CO <sub>2</sub> | Carbon Dioxide                                                            |
| ESG             | Environmental, Social, and Governance                                     |
| ECBC            | Energy Conservation Building Code (India)                                 |

## Appendix A

### Appendix A.1. Included Studies Selection Process



**Figure A1.** Flow diagram for document selection process.

**Table A1.** Quality assessment rubric for included studies.

| Criterion                   | Score 0      | Score 1                                                  |
|-----------------------------|--------------|----------------------------------------------------------|
| Source Type                 | Non-reviewed | Institutional/certification doc<br>Peer-reviewed journal |
| POE Data Availability       | None         | Full pre- and post-occupancy data                        |
| Methodological Transparency | Unclear      | Clear methods, metrics,<br>timeframe                     |
| TBL Relevance               | Not related  | Addresses at least 1 pillar of the<br>TBL framework      |

### Appendix A.2. Performance Gaps: Comparative Table Between Case Studies

**Table A2.** Comparative Table for Performance Gaps Between Case Studies.

| Building        | Expected | Actual | Gap (%) | Score | Comment                     | Key Insight                                                                                                                 |
|-----------------|----------|--------|---------|-------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Bullitt Center  | 100%     | 130%   | −30%    | 2     | Exceeded performance target | Adaptive monitoring enables superior outcomes                                                                               |
| Lawrence Center | 100%     | 70%    | +30%    | 4     | Major under-performance     | Poor commissioning led to inefficiency                                                                                      |
| Bloomberg HQ    | 100%     | 135%   | −35%    | 1     | Strong over-performance     | Recalibration improved performance                                                                                          |
| Delos HQ        | 100%     | 90%    | +10%    | 3     | Slight under-performance    | Real-time environmental controls improved thermal and acoustic conditions but inadvertently increased HVAC energy intensity |

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